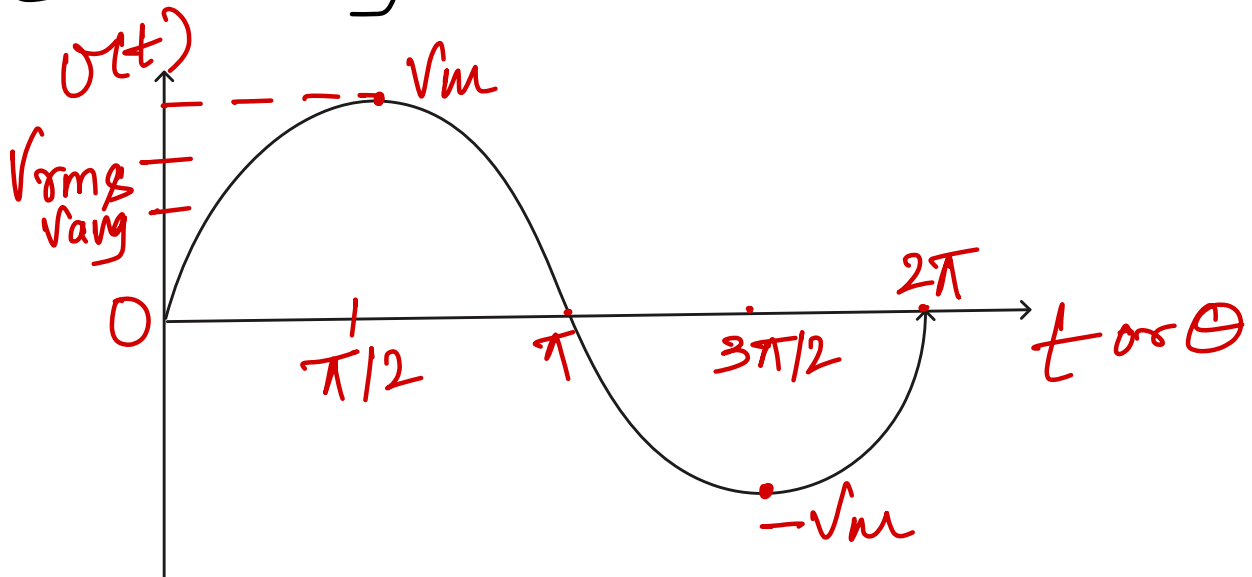


Module $\rightarrow 3$

* Sinusoidal Signal Representation:

$\rightarrow$ A general AC voltage is given by $V(t) = V_m \sin(\omega t)$

A signal whose magnitude varies with time and changes direction periodically



- Peak value - V_m volts
- Angular Frequency ($\omega = 2\pi f$) - rad/s
- Time period (T) = $1/f$ in sec.
time taken in one complete cycle.
- f (frequency) $\rightarrow$ hertz
Number of oscillations per sec.

→ Why sinusoidal?

- generated naturally in alternators
- Mathematical simplicity in analysis.

* Derivation for Average value of a sine wave.

→ Mean of instantaneous values over a cycle.

• for a complete cycle of a sine wave.

$$V_{avg} = 0$$

As, +ve and -ve part of sine wave cancel each other.

→ So, we calculate Avg. value over half cycle,

(i) For 0 to π :

$$V_{avg} = \frac{1}{\pi} \int_0^{\pi} V_m \sin \theta d\theta$$

$$= \frac{1}{\pi} V_m [\cos \theta]_0^{\pi}$$

$$= \frac{1}{\pi} V_m [\cos 0 - \cos \pi]$$

$$V_{avg} = \boxed{\frac{2 V_m}{\pi}}$$

(ii) for 0 to $\pi/2$:

$$\begin{aligned} V_{avg} &= \frac{1}{\pi/2} \int_0^{\pi/2} V_m \sin \theta d\theta \\ &= \frac{2}{\pi} V_m [-\cos \theta]_0^{\pi/2} \\ &= \frac{2}{\pi} V_m [1 - 0] \end{aligned}$$

$$V_{avg} = \frac{2 V_m}{\pi}$$

NOTE: If we consider either $\pi/2$ or π for half wave V_{avg} always $= \frac{2 V_m}{\pi}$
→ generalized Avg. value formulae
For any signal; $V_{avg} = \frac{1}{T} \int_0^T V(t) dt$

* Derivation for RMS (Root Mean Square) value for sine wave:

Def:ⁿ It is a value that, when applied across a resistance produces same amount of heat that a Direct current (DC) voltage of same magnitude would produce.

→ Derivation of generalized RMS Value formulae:

As, RMS (Root Mean Square) is based on, heating effect (Power):

As, Power, $P = \frac{V^2}{R}$

• for, AC - $P_{avg} = \frac{1}{T} \int_0^T \frac{V^2(t)}{R} dt$ - (i)

• for DC, $P = \frac{V_{dc}^2}{R}$ - (ii)

Equating both eqn (i) and (ii),

$$\frac{V_{dc}^2}{R} = \frac{1}{T} \int_0^T \frac{V^2(t)}{R} dt$$

Cancelling R we get,

$$V_{dc}^2 = \frac{1}{T} \int_0^T V^2(t) dt$$

$$\therefore \left[V_{rms} = \sqrt{\frac{1}{T} \int_0^T V^2(t) dt} \right]$$

→ Physical meaning of RMS is that if $V_{rms} = 230V$ of AC it produces same heat as, 230V DC across a Resistor.

• AC Meters shows RMS value not, peak value.

→ RMS Value Derivation for a sine wave:

$$V_{rms} = \sqrt{\frac{1}{\pi/2} \int_0^{\pi/2} (V_m \sin \theta)^2 d\theta}$$

$$= \sqrt{\frac{2}{\pi} V_m^2 \int_0^{\pi/2} \sin^2 \theta d\theta}$$

$$= \frac{\sqrt{2} V_m}{\sqrt{\pi}} \sqrt{\left[\frac{\theta}{2} - \frac{1}{4} \sin 2\theta \right]_0^{\pi/2}}$$

$$= \frac{\sqrt{2} V_m}{\sqrt{\pi}} \sqrt{\left(\frac{\pi}{4} - 0 \right) - (0 - 0)}$$

$$= \frac{\sqrt{2} V_m}{\sqrt{\pi}} \times \frac{\sqrt{\pi}}{2} = \frac{1}{\sqrt{2}} V_m$$

So,

$$\left[V_{rms} = \frac{1}{\sqrt{2}} V_m \approx 0.707 V_m \right]$$

* Form factor :-

Defⁿ - for comparing different waveforms of same frequency and amplitude but of different wave shape a parameter is called as form factor and is defined as ratio of its RMS & Average Value.

$$F.f = \frac{V_{rms}}{V_{avg}}$$

→ It indicates the waveform shape

- Sine wave → 1.11
- Square wave → 1
- Triangular wave → different

→ For a sine wave f.f can be calculated as,

$$f.f = \frac{V_{rms}}{V_{avg}} = \frac{0.707 V_m}{0.637 V_m} = 1.1$$

* Peak factor (PF) or Crest factor:

Defⁿ → Peak factor is defined as the ratio of the maximum (peak) value to the RMS value of an alternating quantity.

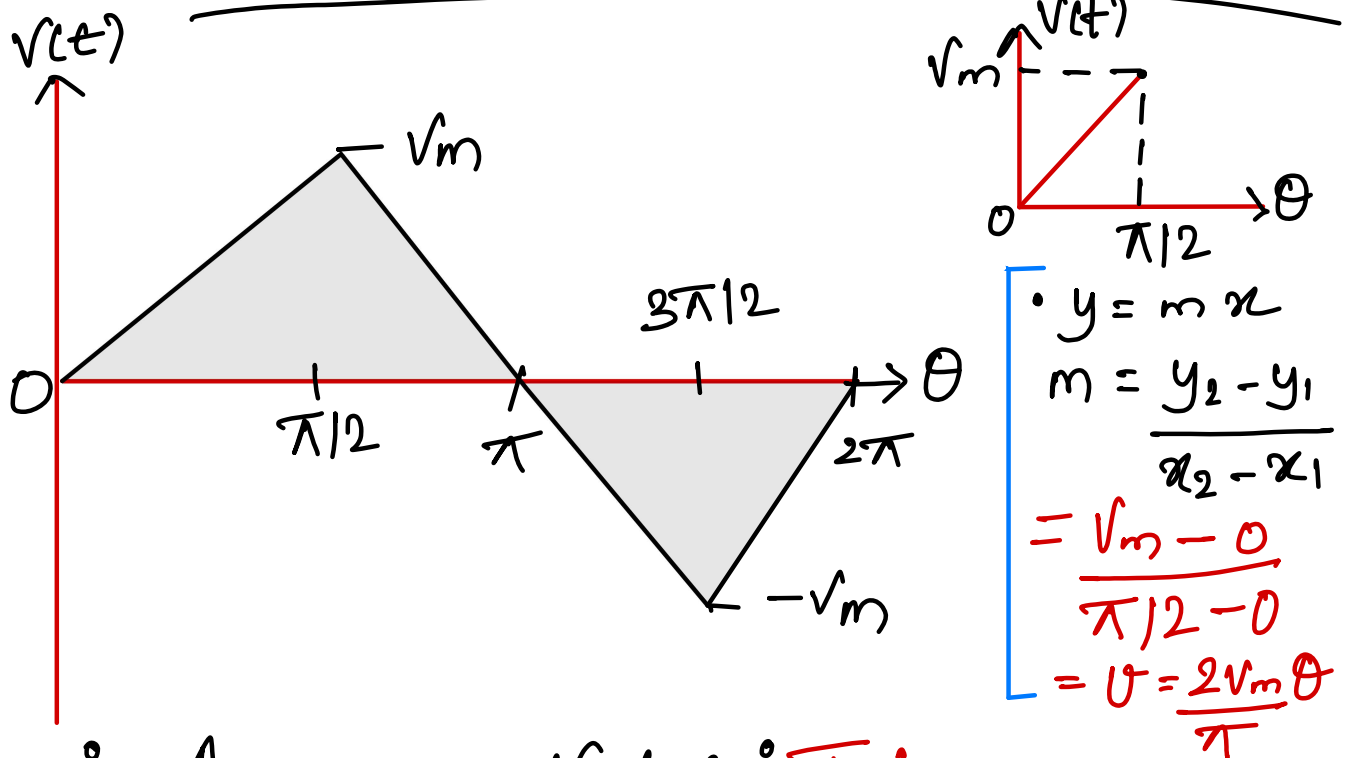
→ It shows how sharp the waveform peak is.

$$\left[\bullet \text{ PF} = \frac{V_m}{V_{\text{rms}}} \right]$$

→ For a sine wave,

$$\left[\text{PF} = \frac{V_m}{\frac{V_m}{\sqrt{2}}} = \sqrt{2} = 1.414 \right]$$

* Triangular Wave (All parameters)



(i) Average Value: Triangular,

$$V_{avg} = \frac{1}{\pi/2} \int_0^{\pi/2} \frac{2v_m\theta}{\pi} d\theta$$

$$= \frac{1}{\pi/2} \times \frac{2}{\pi} v_m \left[\frac{\theta^2}{2} \right]_0^{\pi/2}$$

$$= \frac{4}{\pi^2} v_m \left[\left(\frac{\pi}{2} \right)^2 - 0 \right]$$

$$= \frac{4}{\pi^2} v_m \times \frac{\pi^2}{4 \times 2}$$

$$= \boxed{\frac{v_m}{2} \approx 0.5 v_m}$$

(ii) RMS Value: Triangular wave

$$V_{rms} = \sqrt{\frac{1}{\pi/2} \int_0^{\pi/2} \left(\frac{2V_m}{\pi} \theta \right)^2 d\theta}$$

$$= \sqrt{\frac{2}{\pi} \times \left(\frac{2V_m}{\pi} \right)^2 \int_0^{\pi/2} \theta^2 d\theta}$$

$$= \sqrt{\frac{8V_m^2}{\pi^3} \left[\frac{\theta^3}{3} \right]_0^{\pi/2}} = \frac{2\sqrt{2}V_m}{\pi} \sqrt{\frac{1}{\pi} \left[\frac{(\pi/2)^3}{3} - 0 \right]}$$

$$= \frac{2\sqrt{2}}{\pi} \sqrt{\frac{1}{\pi} \times \frac{\pi^3}{(2)^3 \times 3}} = \frac{2\sqrt{2}}{\pi} \sqrt{\frac{\pi^2}{2^2 \times 3}}$$

$$= \frac{2\sqrt{2}}{\pi} \times \frac{\pi}{2} \times \sqrt{\frac{1}{6}} V_m = \frac{\sqrt{2} \times 1}{\sqrt{2} \times \sqrt{3}} \times V_m$$

$$V_{avg} = \frac{V_m}{\sqrt{3}}$$

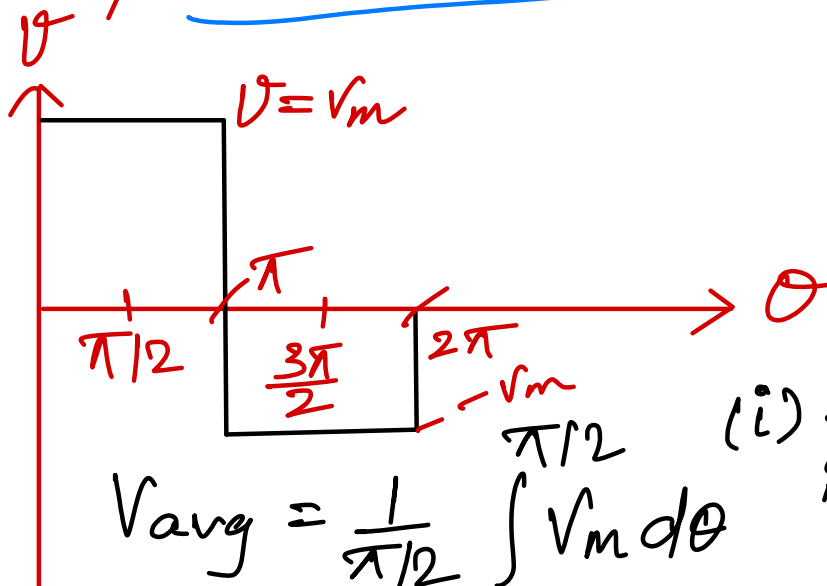
(iii) Form factor (FF): Triangular

$$FF \rightarrow \frac{V_{rms}}{V_{avg}} = \frac{0.577 V_m}{0.5 V_m} = 1.15$$

(iv) Peak factor:

$$\frac{V_m}{V_{rms}} = \frac{V_m}{0.577 V_m} = 1.732$$

* Square Wave: All Parameters



$$V_{avg} = \frac{1}{\pi/2} \int_0^{\pi/2} V_m d\theta \quad \text{(i) Avg. value square}$$

$$= V_m \left[\theta \right]_0^{\pi/2} \times \frac{2}{\pi}$$

$$= \boxed{V_{avg} = V_m}$$

$$(ii) V_{rms} = \sqrt{\frac{1}{\pi/2} \int_0^{\pi/2} V_m^2 d\theta} = \sqrt{V_m^2} = \boxed{V_m}$$

(iii) Form factor (Ff):

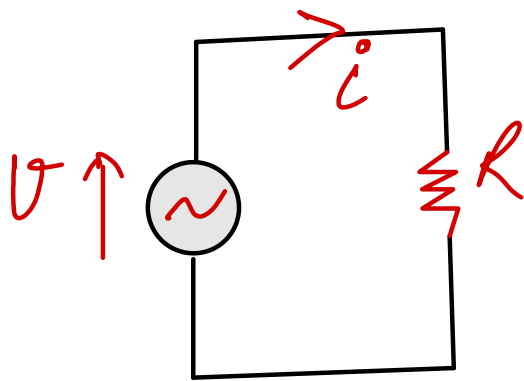
$$ff = \frac{V_{rms}}{V_{avg}} = \frac{V_m}{V_m} = 1$$

(iv) Peak factor:

$$Pf = \frac{V_m}{V_{rms}} = \frac{V_m}{V_m} = 1$$

* Series 1 ϕ (phase) CRTs:

(i) Pure Resistance (R):



Ohm's law,
 $V = RI$

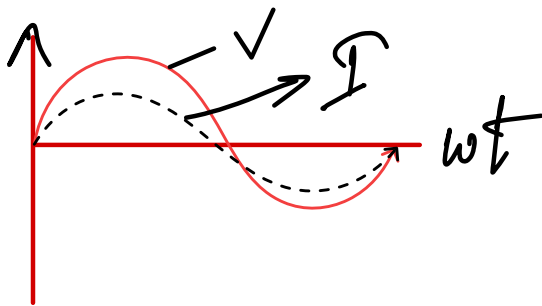
$$i = \frac{V}{R} \text{ (A)}$$

$$V = V_m \sin \omega t \text{ (V)}$$

$$\text{So, } i = \frac{V_m}{R} \sin \omega t \text{ (A)}$$

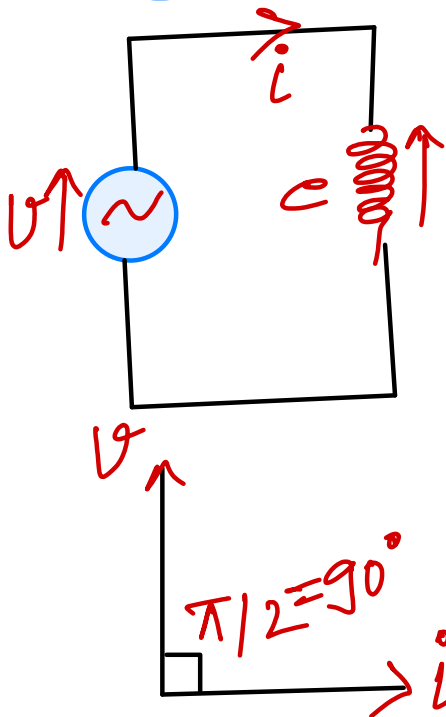
$R \rightarrow I_m$

$$i = I_m \sin \omega t \text{ (A)}$$



$$\phi = 0^\circ \text{ (in phase)}$$

(ii) Pure Inductance:



$$\psi = Li$$

So, induced emf

$$= e = \frac{d\psi}{dt} = -L \frac{di}{dt} \text{ Volts}$$

'e' will have its max value when the rate of change of current, $\frac{di}{dt}$ is at its maximum values.

• In pure inductor voltage leads by 90° .

→ Average power across inductor is zero, so, for a perfect inductor dissipates zero power.

→ $e = -L \frac{di}{dt}$; $i = I_m \sin \omega t$
Lenz law for polarity

→ $V = L \frac{di}{dt} = L \frac{d(I_m \sin \omega t)}{dt}$

$V = \omega L I_m \cos \omega t$

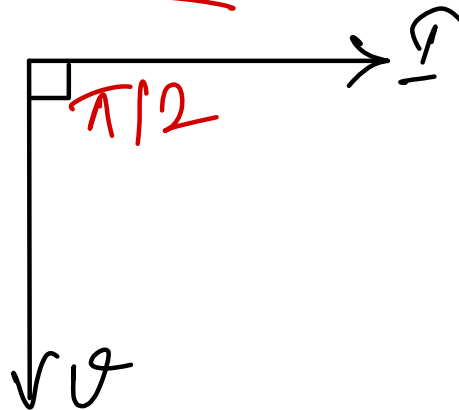
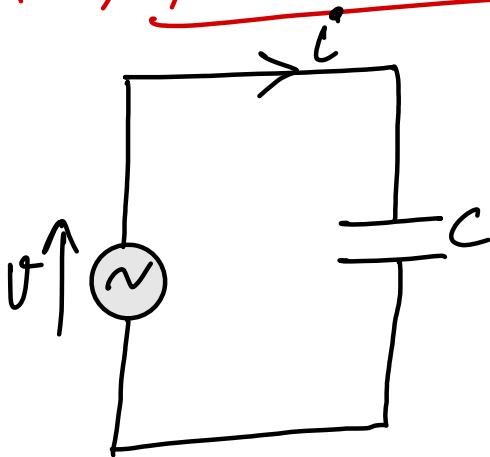
So, at $t = 0$, $V = V_m$; $\cos \omega t = 1$

$V_m = \omega L I_m \Rightarrow \frac{V_m}{I_m} = \omega L \Omega$

So, $X_L = \omega L$



(iii) Pure capacitor:-



$q = C V$; $i = \frac{dq}{dt}$

$\left[i = C \frac{dv}{dt} \right]$

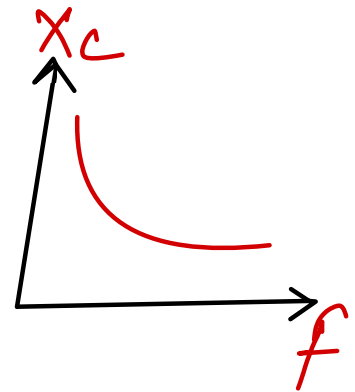
; $V = V_m \sin \omega t$

$i = \omega C V_m \sin \omega t$

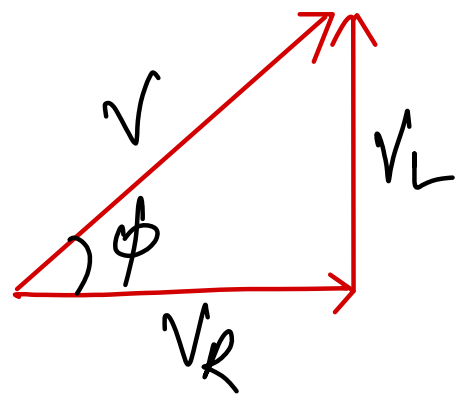
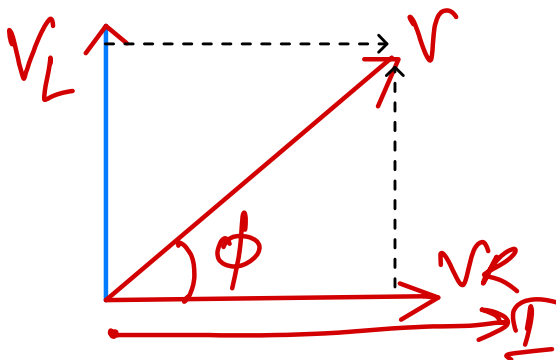
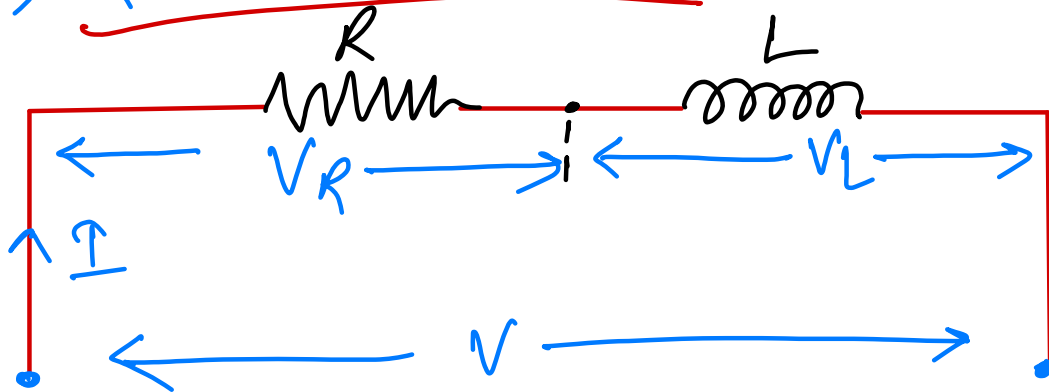
At $t=0$; $i = I_m$; $\cos \omega t = 1$,

$$\rightarrow \frac{V_m}{I_m} = \frac{V}{I} = \frac{1}{\omega C}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$



(iv) RL in series:



$$V^2 = V_R^2 + V_L^2$$

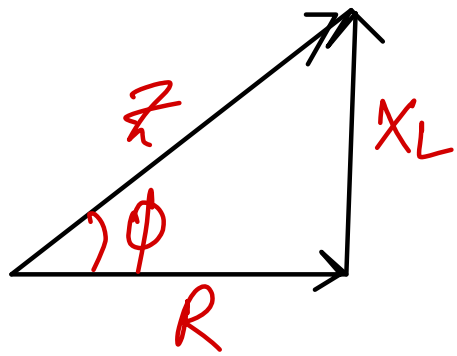
where,

$$V_R = IR; \quad V_L = IX_L; \quad V = IZ$$

$\Rightarrow$ By Pythagoras theorem,

$$(IZ)^2 = (IR)^2 + (IX_L)^2$$

$\Rightarrow [Z = \sqrt{R^2 + X_L^2}] \rightarrow$ impedance in RL ckt.

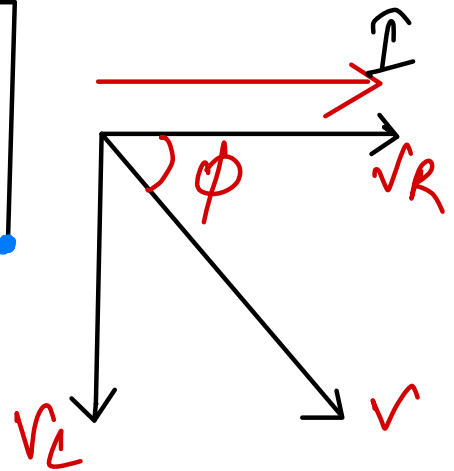
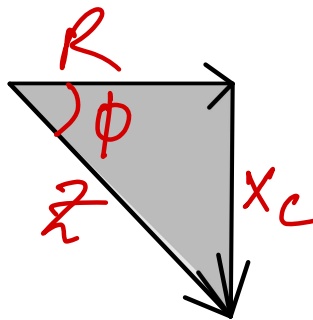
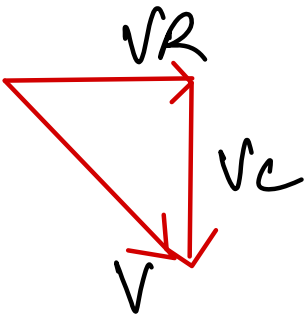
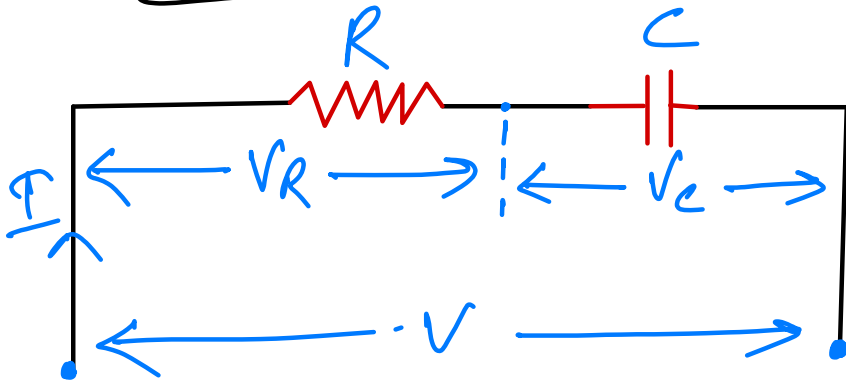


$\therefore \cos \phi = \frac{R}{Z} = \frac{V_R}{V}$

$\therefore \sin \phi = \frac{X_L}{Z} = \frac{V_L}{V}$

$\therefore \tan \phi = \frac{X_L}{R} = \frac{V_L}{V_R}$

(v) RC in series:

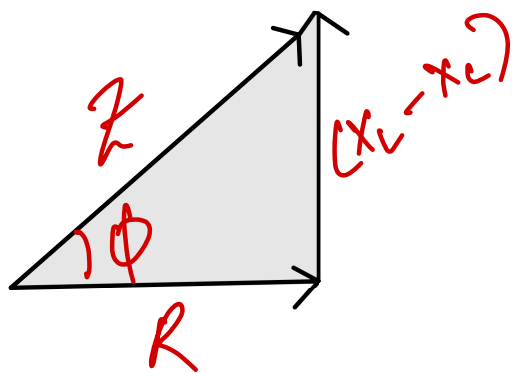
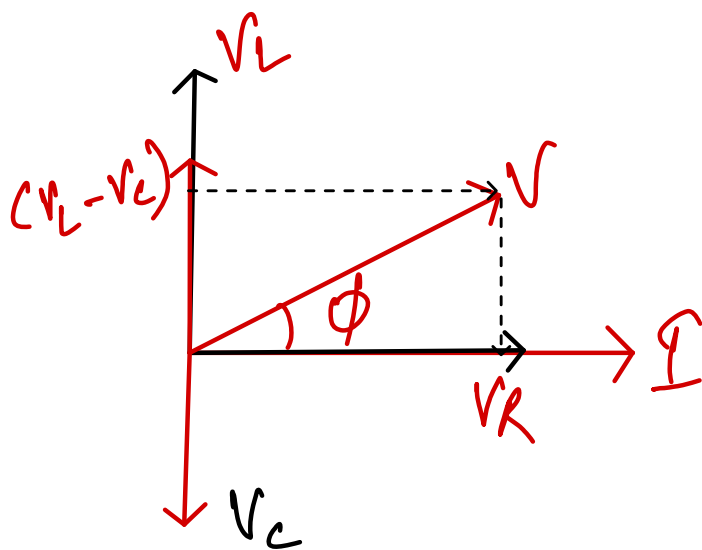
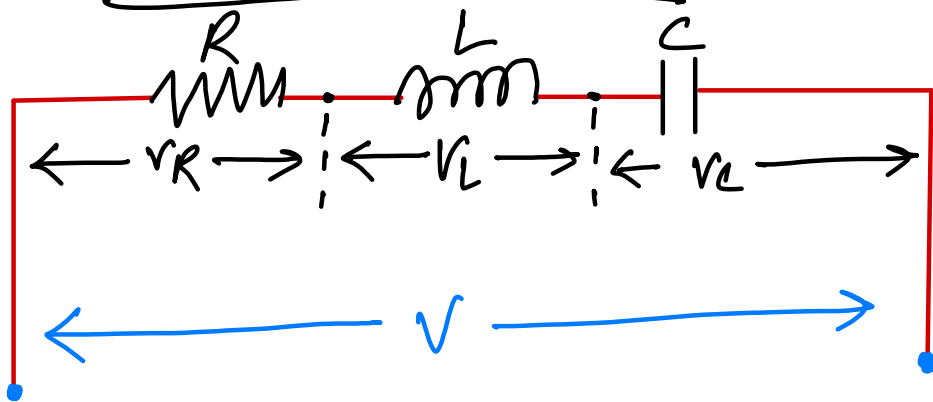


So, $Z = \sqrt{R^2 + X_C^2}$; $\phi = \cos^{-1}(\frac{R}{Z})$

$\therefore \sin \phi = \frac{X_C}{Z} = \frac{V_C}{V}$

$\therefore \tan \phi = \frac{X_C}{R} = \frac{V_C}{V_R}$

(vi) RLC in series:



So,

$$V^2 = V_R^2 + (V_L - V_C)^2$$

$$\Rightarrow \underline{Z = \sqrt{R^2 + (X_L - X_C)^2}}$$

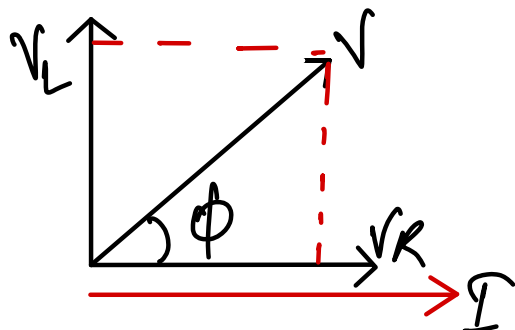
$$\bullet \cos \phi = \frac{R}{Z}$$

$$\bullet \sin \phi = \frac{X_L - X_C}{Z}$$

$$\bullet \tan \phi = \frac{X_L - X_C}{R}$$

* Power in 1- ϕ AC circuits:

Power is dissipated only by the resistance in the circuit. Pure L & C does not dissipate any power.



$$P = V_R I;$$

from the phasor

$$\bullet V_R = V \cos \phi$$

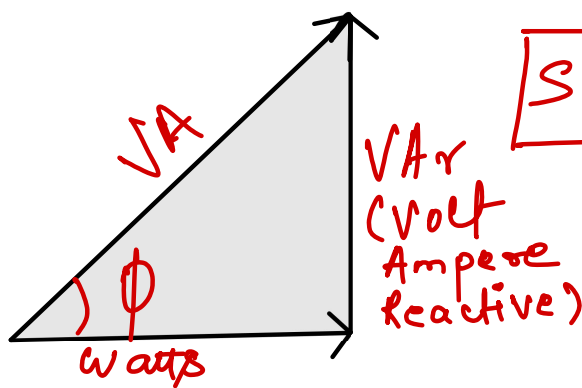
So,

$$P = VI \cos \phi$$

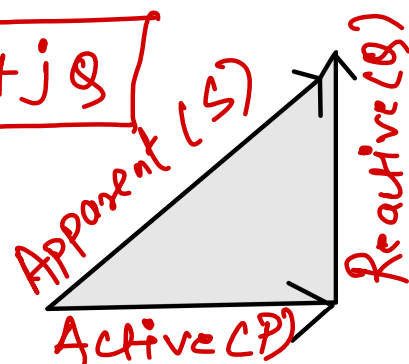
Power factor:

Ratio of true power to the product of RMS values of applied voltage and current (Apparent Power).

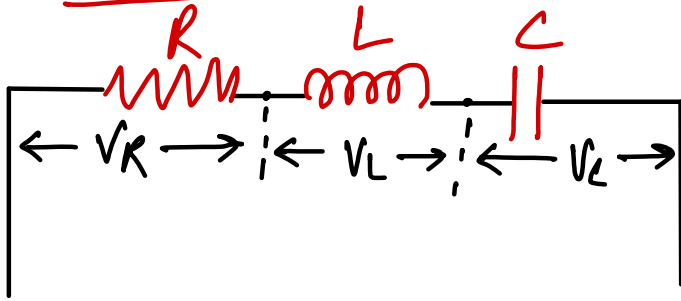
$$\text{Power factor} = \frac{VI \cos \phi}{VI} = \cos \phi$$



$$S = P + jQ$$



Series Resonance:



* Keeping variable frequency (f):

① When frequency is low,

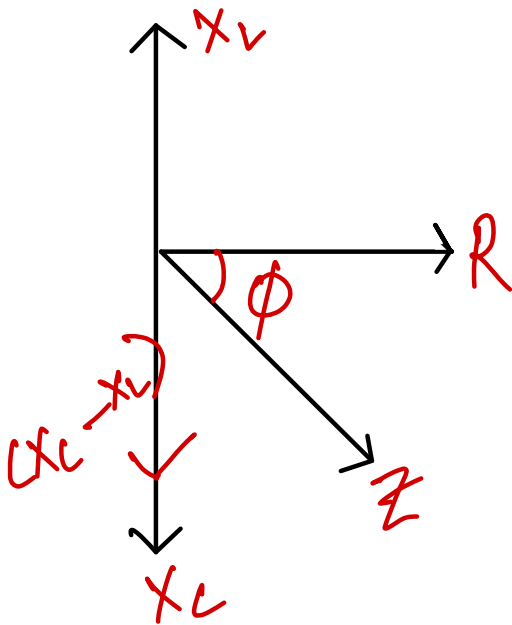
$$X_C > X_L$$

② freq. is high

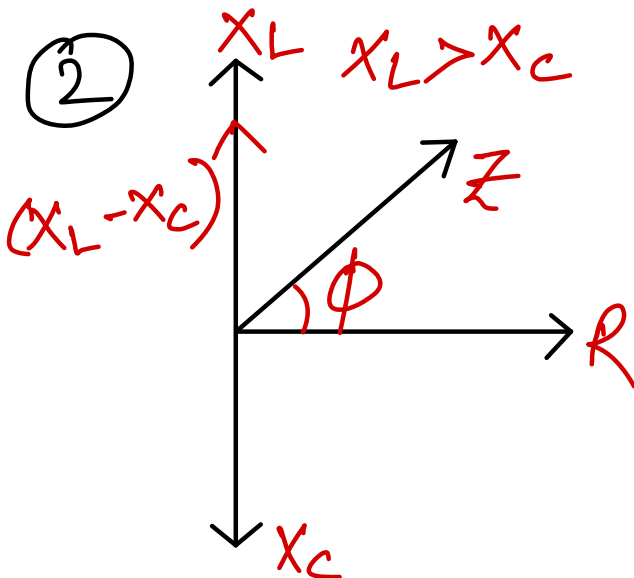
$$X_C < X_L$$

③ $X_C = X_L$
(Resonance condition)

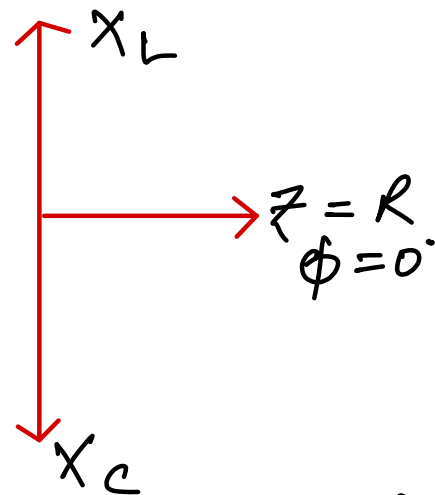
① $X_C > X_L$ (Phasor):



② $X_L > X_C$



③ $X_L = X_C$ (Resonance cond')



So, $Z = R$ (low impedance)

$$2\pi fL = \frac{1}{2\pi fC}$$

$$\Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Resonating frequency

* Series RLC Plots:

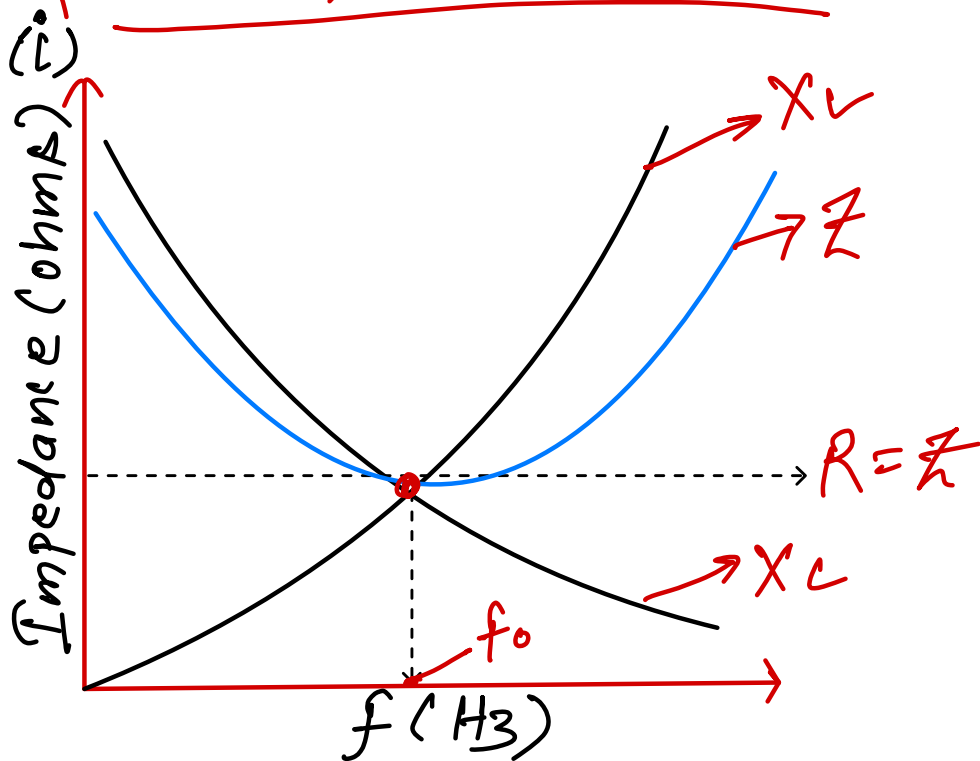
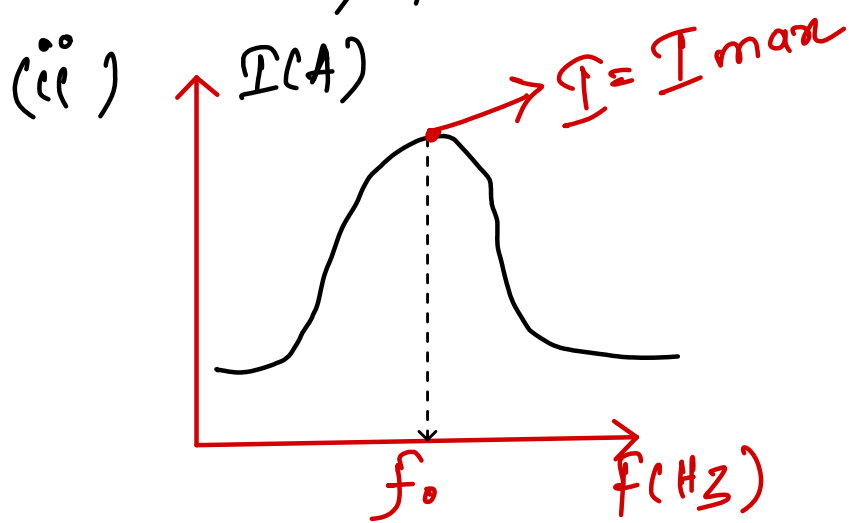


Fig: Z vs f curve



→ Q (Quality) factor indicates how easily current flows at the resonant frequency and reflects the ckt transient response quality. A high ' Q ' factor means the ckt responds sharply to the selected frequency

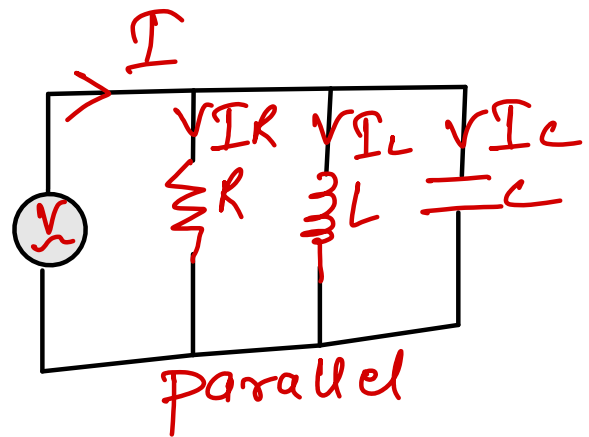
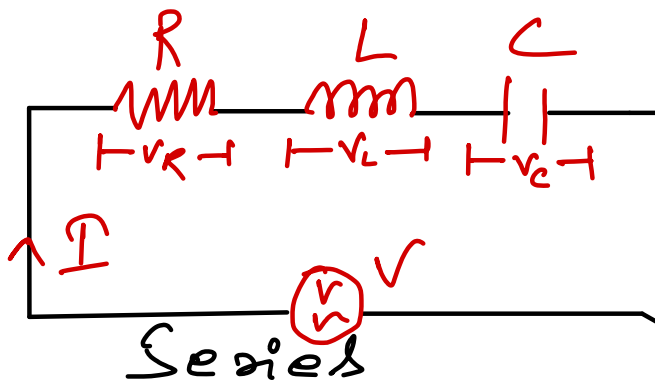
$$Q = \frac{X_L}{R} = \frac{2\pi f_0 L}{R} = \boxed{\frac{X_C}{R} = \frac{1}{2\pi f_0 C R}}$$

NOTE: Smaller the resistance, $R \rightarrow$
Higher the quality factor, Q , and the
sharper the resonance characteristics
become,

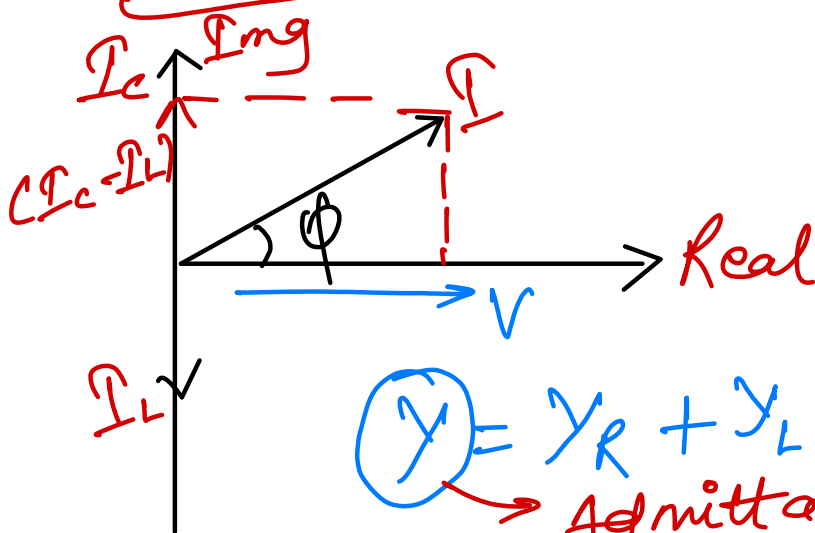
$$Q = \frac{2\pi f_0 L}{R} = \frac{1}{2\pi f_0 C R} = \frac{1 \times 2\pi \times \sqrt{LC}}{2\pi \times 1 \times C R}$$

$$\boxed{Q = \frac{1}{R} \sqrt{\frac{L}{C}}}$$

A RLC CKT:



Parallel RLC CKT:



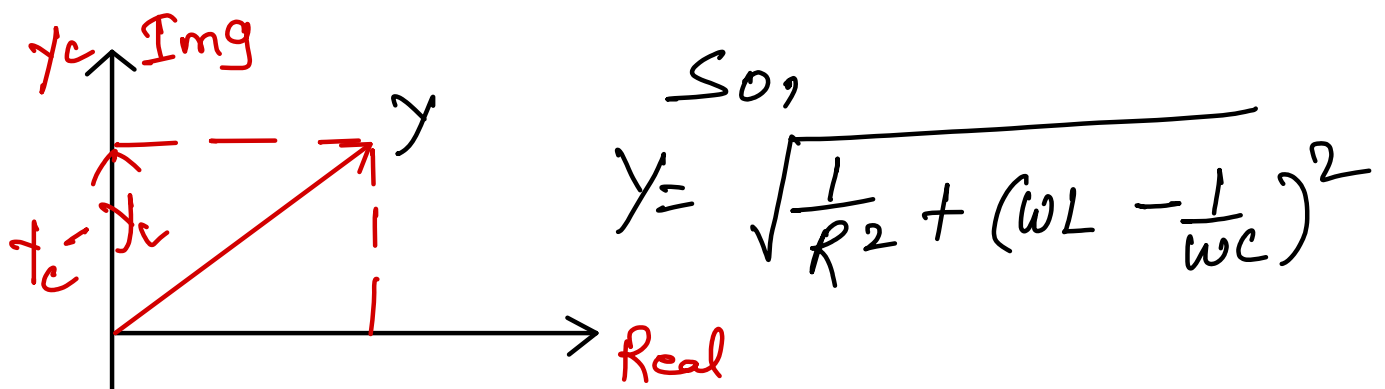
$$Y = Y_R + Y_L + Y_C$$

→ Admittance is the inverse of the impedance.

$$= \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

$$= |Y| = \sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

$$\text{and, } |Z| = \frac{1}{|Y|} = \frac{1}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$



At, resonant frequency, ' f_0 '
the condⁿ would be,

$$2\pi fC - \frac{1}{2\pi fL} = 0$$

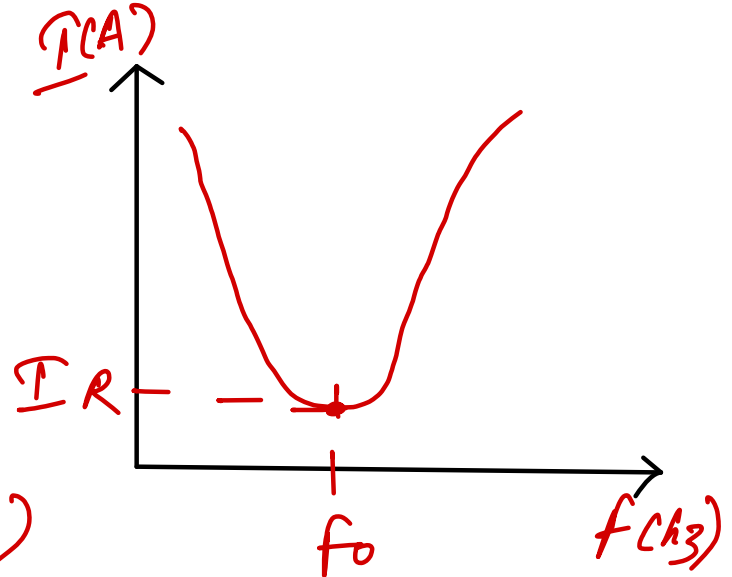
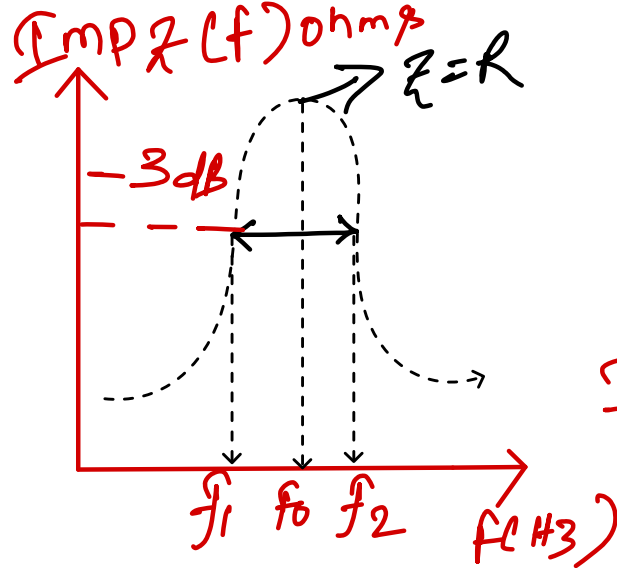
$$\Rightarrow 2\pi fC = \frac{1}{2\pi fL}$$

$$\Rightarrow f^2 = \frac{1}{(2\pi)^2 LC}$$

$$\Rightarrow f_0 = \frac{1}{2\pi \sqrt{LC}} \quad \rightarrow \text{same as Series}$$

NOTE:

- ① Absolute value of the impedance $|Z|$ becomes max. at the resonant frequency.
- ② At the resonant state, absolute value of the current flowing through the ckt ($|I|$) becomes minimal.



* Bandwidth indicates the range of frequencies over which the Ckt. functions effectively.

$$BW = \Delta f = f_2 - f_1 \text{ OR } BW = \frac{f_0}{Q}$$

→ ω_1 and ω_2 are frequencies at which the impedance is half its max. value centered around resonant frequency.

$$\rightarrow Q \text{ factor (parallel)} \rightarrow \frac{R}{2\pi fL} = 2\pi fCR = R \sqrt{\frac{C}{L}}$$

Series ckt.

$$\cdot |Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\cdot I = I_R = I_L = I_C$$

$$\cdot V = V_R + V_L + V_C$$

$$\cdot Q = \frac{2\pi f_0 L}{R} = \frac{1}{2\pi f_0 C R}$$
$$= \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$\cdot BW = \frac{f_0}{Q} = \frac{1}{2\pi C R}$$

Parallel ckt.

$$\cdot |Z| = \frac{1}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

$$\cdot I = I_R + I_L + I_C$$

$$\cdot V = V_R = V_L = V_C$$

$$\cdot Q = \frac{R}{2\pi f_0 L} = 2\pi f_0 C R$$
$$= R \sqrt{\frac{C}{L}}$$

$$\cdot BW = \frac{f_0}{Q} = \frac{R}{2\pi L}$$

→ ↙
Comparison